S^3Net: Spectral–Spatial–Semantic Network for Hyperspectral Image Classification With the Multiway Attention Mechanism

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***Abstract*— In hyperspectral image (HSI) classification, it is a great challenge on how to extract key informative spectral–spatial features efficiently and suppress useless features from abundant spectral–spatial information. In this article, inspired by the attention mechanism of the human visual system, we propose a novel spectral–spatial–semantic network (S^3Net) with the multiway attention mechanism for HSI classification. The S^3Net consists of a spectral branch, a spatial branch, and a multiscale semantic module. The spectral branch extracts the multiway spectral features with a dense spectral block and a multiway spectral attention module. The spatial branch extracts the multiway spatial features with a multiscale spatial block and a multiway spatial attention module. The multiscale semantic module extracts spectral–spatial–semantic features that are used for classification. In the proposed S^3Net, the multiway attention modules in spectral and spatial branches are built to enhance the extraction ability of the key informative spectral–spatial features. The multiscale spatial block in the spatial branch is designed to learn strong complementary and related information. The Res2Net in the multiscale semantic module is used to learn multiscale semantic features at a granular level. A large number of experimental results demonstrate that, on the University of Pavia, Kennedy Space Center, and Pavia Center data sets, the proposed S^3Net achieves higher classification accuracy than state-of-the-art methods on the limited training samples. Remarkably, our S^3Net also achieves the best performance on the mineral exploration HSI data set, called Huoshaoyun, which is collected by the GaoFen-5 (GF-5) satellite.**

***Index Terms*— Huoshaoyun data set, hyperspectral image (HSI) classification, multiway spatial attention, multiway spectral attention.**

I. Introduction

HYPERSPECTRAL image (HSI) is a special type of remote image that has abundant spectral–spatial information. HSI has been studied in many fields, including environment management [1], mineral exploitation [2], and anomaly detection [3]. Especially, HSI classification is one of the important tasks in the analysis of HSI, which assigns a category label to each pixel. In the past decade, HSI classification has been studied by more and more researchers [4]–[19].

Traditional machine learning methods [4]–[9], [12]–[14], [20] mainly adopt the following two steps: feature extraction and classifier training. There are many techniques to achieve feature extraction, such as feature mining techniques [21] and hidden Markov random field [6], which reduces the high dimensionality of HSI and extracts the discriminative features or bands. Then, the well-designed classifiers are trained with the discriminative features and further predict the HSI classification probability. However, these traditional machine learning methods usually suffer from low classification accuracy due to poor feature representation and generalization ability.

With the development of computer vision, to further extract more discriminative features, deep learning-based methods have achieved great success in the HSI classification task due to the strong feature representation and generalization ability [10], [11], [15], [16], [18], [19], [22], [23]. For example, Chen *et al.* [11] first proposed a deep learning framework (stacked autoencoders) to obtain useful high-level features. Then, the classifiers built in this deep learning framework can achieve higher accuracy than most traditional methods. Li *et al.* [24] further proposed a fully convolutional neural network (CNN) containing convolution, deconvolution, and pooling layers to learn the enhanced deep features. Then, the optimized extreme learning machine (ELM) was used for HSI classification. Cao *et al.* [23] adopted a CNN to learn the posterior class distributions. Acquarelli *et al.* [25] used the CNN and data augmentation for HSI classification. Then, a tailored loss function penalizing differences among weights of the network was used to train the network. Although these deep learning-based HSI classification methods can extract deep features, it is still difficult for them to pay more attention to some key informative features and suppress useless features from abundant spectral–spatial information. Therefore, how to find and focus on the key informative features of HSI is an essential problem.

Inspired by the visual attention mechanism of human visual perception, many researchers integrate the attention mechanism as an effective technique into visual tasks, such as

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object detection [26], image caption [27], and video classification [28]. Since the attention mechanism can focus on specific parts of the entire image and heighten the sensitivity to features containing the most valuable information [29], [30], some researchers have also applied the attention mechanism to HSI classification [31]–[34]. Some state-of-the-art methods are also proposed, such as spectral–spatial attention network (SSAN) [32], double-branch multiattention (DBMA) mechanism network [33], and double-branch dual-attention (DBDA) mechanism network [34]. In detail, Mei *et al.* [32] proposed an SSAN. Ma *et al.* [33] proposed a DBMA mechanism network. Li *et al.* [34] proposed a DBDA mechanism network. Although the three networks can obtain good classification performance, their attention mechanisms are too simple to learn the key informative spectral features and spatial features. Meanwhile, it is difficult for them to learn high-level semantic features due to their simple concatenate operation between spectral features and spatial features.

Usually, semantic representation of an image can be regarded as the probability of an image belonging to each class. In the HSI classification, semantic representation of one pixel in HSI can also be regarded as the probability of this pixel belonging to each class. Zhang *et al.* [35] used the probability the support vector machine (SVM) to map multiple features from different low-level subspace to the same semantic space (probability space). Huang and Zhang [36] also integrated the spectral, spatial, and semantic features and proposed a new multifeature model to improve the classification accuracy. Zou *et al.* [37] proposed a spectral–spatial 3-D fully convolutional network to jointly explore the spectral–spatial information and the semantic information. This network first used the semantic segmentation method to generate the rough HSI pixel classification results. The 3-D branch then was used to extract the spectral–spatial features. Yu *et al.* [18] first used the hash learning method to extract the hash semantic features. Then, a simple convolution neural network was utilized to explore the classification features that were used for classification.

In this article, we propose a novel spectral–spatial–semantic network (S^3Net) with the multiway attention mechanism for HSI classification. The S^3Net contains the spectral branch that extracts multiway spectral features and the spatial branch that extracts multiway spatial features. The spectral branch includes a dense spectral block and a multiway spectral attention module. The spatial branch includes a multiscale spatial block and a multiway spatial attention module. Two branches can reduce the interference between the multiway spectral features and multiway spatial features. To integrate two types of features, a new cross-fusion operation is designed to generate spectral–spatial features. Finally, we use the multiscale semantic module to map low-level spectral/spatial features to the high-level spectral–spatial–semantic features that are used for classification. The proposed S^3Net contains the following four contributions.

1) We propose a novel S^3Net with the multiway attention mechanism for HSI classification, which includes a spectral branch, a spatial branch, and a multiscale semantic module. It can extract more sufficient and various spectral–spatial–semantic features.
2) We apply both the multiway spectral attention and the multiway spatial attention in the S^3Net. The multiway spectral attention aims at emphasizing informative spectral features while suppressing useless spectral features. The multiway spatial attention puts more emphasis on the informative areas in the spatial features.
3) We design a new multiscale spatial block to learn the strong complementary and related information from the spatial branch.
4) To make full use of the multiway spectral features and the multiway spatial features, the multiscale semantic module is proposed to extract the spectral–spatial–semantic features that are used for classification.

The experimental results on the University of Pavia (UP), Kennedy Space Center (KSC), and Pavia Center (PC) data sets demonstrate that the S^3Net achieves higher classification accuracy than state-of-the-art methods on the limited training samples. Our method is also utilized in a Huoshaoyun HSI data set that is used for mineral exploration. It is collected by the GaoFen-5 (GF-5) satellite in the Huoshaoyun area of Xinjiang province, China. The experimental results show that our method successfully divides the distribution of minerals in the Huoshaoyun area. Especially, our proposed S^3Net also achieves the best performance.

The remainder of this article is organized as follows. In Section II, we illustrate some related and state-of-the-art works. In Section III, we present the S^3Net framework. In Section IV, we introduce three widely used HSI data sets and a new Huoshaoyun HSI data set for mineral exploration. Furthermore, we design extensive experiments to evaluate the HSI classification ability of the S^3Net framework. Finally, this article is concluded in Section V.

II. Related Work

In the HSI classification task, there are many existing machine learning methods to solve the HSI classification problem [4]–[9], [12]–[14], [20], [38]. Duan *et al.* [12] proposed a local manifold-based sparse discriminant learning (LMSDL) feature extraction method based on the manifold learning and sparse representation, which can reveal the local manifold-based sparse structure of data. They construct two geometrical sparse graphs to represent the discriminant relationship between samples and the geometrical and sparse neighbors. Luo *et al.* [13] proposed a sparse-adaptive hypergraph discriminant analysis (SAHDA) method to obtain the embedding features of the HSI. It uses sparse representation to reveal the structural relationships of the HSI adaptively. Duan *et al.* [14] proposed a novel dimensionality reduction method named local constraint-based sparse manifold hypergraph learning (LC-SMHL) algorithm to discover the manifold-based sparse structure and the multivariate discriminant sparse relationship of HSI. LC-SMHL learns an optimal projection for mapping data into low-dimensional space in which embedding features not only discover the manifold structure and sparse relationship of original data but also possess strong discriminant power for HSI classification. Luo *et al.* [20] proposed a spatial–spectral hypergraph discriminant analysis (SSHGDA). It constructs a reconstruction between-class

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scatter matrix, a weighted within-class scatter matrix, an intraclass spatial–spectral hypergraph, and an interclass spatial–spectral hypergraph to represent the intrinsic properties of HSI. Li *et al.* [38] proposed a framework that utilized local binary patterns (LBPs) to extract local image features, and the efficient ELM was used as the classifier. Li *et al.* [8] proposed a discontinuity preserving relaxation strategy for preprocessing of the original HSI and postprocessing of class probability estimates. Especially, they used an iterative relaxation method. Yu *et al.* [9] proposed an iterative target-constrained interference-minimized classifier (TCIMC). They used a stopping rule to decide whether to use Gaussian-filtered TCIMC to further process the classification map. Deng *et al.* [39] proposed a general framework with microtexture. The local response pattern is extended to represent HSI with texture enhancement. The discriminated locality preserving projection is used to reduce the dimensionality of data.

The above methods generally extract hand-crafted features and have difficulty extracting deep discriminative features. Therefore, they suffer from low classification accuracy. With the development of machine learning, the deep learning-based methods have been introduced into HSI classification [11], [15]–[19], [22], [23], [40] and can extract deeper spectral and spatial features. Zhong *et al.* [15] proposed an end-to-end spectral–spatial residual network (SSRN) for HSI classification, whose residual blocks could extract deep spectral and spatial features. Batch normalization (BN) is used to regularize the learning process. Zhong *et al.* [40] also proposed a generative adversarial network and conditional random field (GAN-CRF)-based framework to address the HSI classification problem. The semisupervised framework leverages the merits of discriminative and generative models through a game-theoretical approach. Yu *et al.* [18] proposed a CNN system embedded with an extracted hashing feature for HSI classification that utilized the semantic information of HSI. Especially, the hash functions were built to enhance the presentation of the locality and discriminability of classes. However, this kind of structure was not an end-to-end method that led to a more complicated training process. Zhu *et al.* [19] proposed to use the GAN to perform the HSI classification. The generative CNN tried to generate fake inputs that are as real as possible, and the discriminative CNN tried to classify the real and fake inputs. Zhang *et al.* [22] proposed a diverse region-based CNN to learn contextual interactional features. Wang *et al.* [41] proposed an end-to-end fast dense spectral–spatial convolution (FDSSC) framework for HSI classification. Densely connected structures can extract deep features. The dynamic learning rate, parametric rectified linear units, and dropout layers can reduce the training time. Paoletti *et al.* [16] proposed a new deep CNN to improve the spectral–spatial features uncovered by the convolutional filters of the network. This framework used the pyramid residual-based approach gradually to increase the feature map dimension for higher classification ability. Wang *et al.* [42] further proposed an end-to-end alternately updated spectral–spatial convolutional (AUSSC) network with a recurrent feedback structure for HSI classification, which can refine spectral and spatial features many times under fixed parameters and improve the classification accuracy. Although these deep learning-based HSI classification methods can extract deep features, it is still difficult for them to pay more attention to some key informative features and suppress useless features from abundant spectral–spatial information.

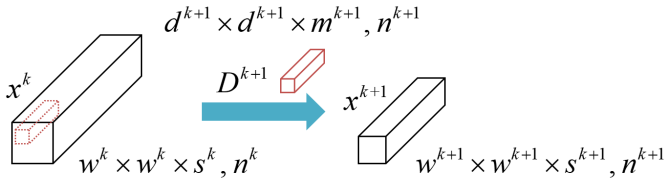


Fig. 1. 3-D convolutional layer. The $(k+1)$th 3-D convolutional layer D^{k+1} conducts convolutional operation for input 3-D feature cubes x^k and generates output 3-D feature cubes x^{k+1}.

Inspired by the human visual perception process [43], the attention mechanism-based methods have been applied in many applications, such as image caption [44] and image question answering [45]. Recently, some attention mechanism-based HSI classification methods have been proposed [32], [34], [46]. Mei *et al.* [32] proposed an SSAN. RNN with attention can learn inner spectral correlations within a continuous spectrum, while CNN with attention can focus on saliency features and spatial relevance between neighboring pixels in the spatial dimension. However, such an RNN structure needs to take a lot of time because of the serial vector input. Fang *et al.* [46] proposed a network using 3-D CNN with spectralwise attention mechanism [multi scale dense connectivity network with spectral-wise attention mechanism (MSDN-SA)]. However, it only considers spectralwise attention while not considering spatialwise attention. Li *et al.* [34] proposed a DBDA mechanism network. Although the DBDA achieved good results, its scales of the spectral attention block and spatial attention block were fixed and had difficulty extracting different scale spectral features and spatial features.

To solve the above problems well, we propose a novel S^{3}Net with the multiway attention mechanism for HSI classification. It consists of a spectral branch, a spatial branch, and a multiscale semantic module. Especially, the 3-D convolutional operation, the DenseNet, and the multiscale Res2Net are the basic elements in the S^{3}Net. We will introduce them in the following paragraphs.

A. 3-D Convolutional Operation

As shown in Fig. 1, we use the kth x^k feature cubes of size $w^k \times w^k \times s^k, n^k$ as input. The $(k+1)$th 3-D convolutional layer D^{k+1} contains n^{k+1} convolutional filters of size $d^{k+1} \times d^{k+1} \times m^{k+1}$, and the subsampling strides of (s_1, s_1, s_2) are used for the convolutional operation. This 3-D convolutional layer can generate the n^{k+1} 3-D feature cubes of size $w^{k+1} \times w^{k+1} \times s^{k+1}$. In detail, the output spatial width is $w^{k+1} = [1 + (w^k - d^{k+1})/s_1]$, and the output spectral depth is $s^{k+1} = [1 + (s^k - m^{k+1})/s_2]$. The BN [47] is used to normalize the input 3-D feature cubes for faster gradient convergence.

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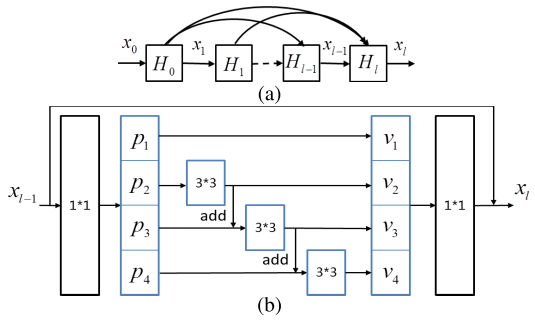


Fig. 2. Figures represent (a) DenseNet and (b) Res2Net architectures separately.

The 3-D convolutional operation can be expressed by

$$x_i^{k+1} = \mathrm{ReLU}\left(\sum_{q=1}^{n^k} \mathrm{BN}(x_q^k) * D_i^{k+1} + b_i^{k+1}\right) \tag{1}$$

where ReLU(.) represents the rectified linear unit activation function [48] that sets elements with negative numbers to zero. x_q^k represents the qth input feature map of the x^k. * represents the 3-D convolutional operation. D_i^{k+1} and b_i^{k+1} denote the parameters and bias of the ith convolutional filter in the $(k+1)$th layer.

B. DenseNet

The densely connection network (DenseNet) connects all layers directly to ensure maximum information flow between each layer of the network [49], which combines features via the concatenate operation. In DenseNet [see Fig. 2(a)], all previous feature maps of l blocks can be used to compute the output of the lth block

$$x_l = H_l\left[x_0, x_1, \ldots, x_{l-1}\right] \tag{2}$$

where H_l is a dense block including convolution layers, activation layers, and BatchNorm layers. The symbols $x_0, x_1, \ldots, x_{l-1}$ represent the feature maps generated by the preceding dense blocks. Especially, DenseNet with L layers have $L(L+1)/2$ connections, while the traditional CNN with equal layers only have L connections.

C. Multiscale Res2Net

Lately, a novel CNN feature extraction network, namely, Res2Net [50] has been proposed for many numerous computer vision tasks, such as image classification [51], object detection [52], semantic segmentation [53], and salient object detection [54]. The Res2Net constructed a hierarchical residual-like connection within one single residual block [55]. This architecture can represent multiscale features at a granular level rather than the previous layerwise manner [56] and regionwise manner [57], which can increase the range of receptive fields for each network layer.

As shown in Fig. 2(b), after the 1×1 convolution layer, the feature maps were equally divided into scale $a = 4$ feature map subsets, denoted by p_i, where $i \in \{1, 2, \ldots, a\}$. Except for p_1, each p_i has a corresponding 3×3 convolution layer, denoted by $R_i(.)$. v_i represents the output of $R_i(.)$. What is more, p_i is added with the output of $R_{i-1}(.)$ and then fed into $R_i(.)$. Therefore, v_i can be expressed as

$$v_i = \begin{cases} p_i & i = 1 \\ R_i(p_i) & i = 2 \\ R_i(p_i + v_{i-1}) & 2 < i \le a. \end{cases} \tag{3}$$

In this process, the 3×3 convolutional operator $R_i(.)$ obtains feature information from all feature splits $\{p_k, k \le i\}$. Then, all v_i's are concatenated to form new feature maps for subsequent 1×1 convolutional operation. Due to the hierarchical residual-like connections in one single residual block, the output X_l feature maps of the Res2Net block contain different receptive field sizes and scales. Inspired by the Res2Net, we build a multiscale semantic module to extract spectral–spatial–semantic features.

III. Proposed Framework

In this section, we will introduce S^{3}Net with the multiway attention mechanism (see Fig. 3), which includes a spectral branch, a spatial branch, and a multiscale semantic module. The spectral branch extracts the multiway spectral features with a dense spectral block and a multiway spectral attention module. The spatial branch extracts the multiway spatial features with a multiscale spatial block and a multiway spatial attention module. The multiscale semantic module extracts spectral–spatial–semantic features that are used for classification.

A. Spectral Branch Structure With the Multiway Spectral Attention Module

The spectral branch (see Fig. 3) contains two 3-D convolutional operations (Conv1, Conv2), a dense spectral block, and a multiway spectral attention module. We take the 3-D cube $9 \times 9 \times 176$ from the UP data as an example to explain the spectral branch.

In the spectral branch, we first use a 3-D convolutional layer (Conv1) with the $1 \times 1 \times 7$ kernel size and the stride (1, 1, 2) to reduce the number of bands of the input 3-D cube, which can capture the reduced feature maps with the shape of $9 \times 9 \times 85, 24$.

Then, we construct a dense spectral block to obtain the dense spectral features. Each 3-D convolution layer (D^{dspe}) in the dense spectral block has 12 channels with a $1 \times 1 \times 7$ kernel size. Finally, we obtain the dense spectral feature maps with the size of $9 \times 9 \times 85, 60$. To further reduce the number of bands, we also use a 3-D convolutional layer (Conv2) with the $1 \times 1 \times 85$ kernel size and 60 channels to generate the spectral feature maps.

However, the 60 spectral feature maps make different contributions to the classification. We build a novel multiway spectral attention module containing m way spectral attention blocks to refine these spectral features (see Fig. 4). First, the input 3-D feature cube $X \in R^{9\times9\times1,60}$ is replicated into m copies: $X^1, X^2, \ldots, X^m$. Second, each copy goes through the spectral attention block to generate the spectral attention maps.

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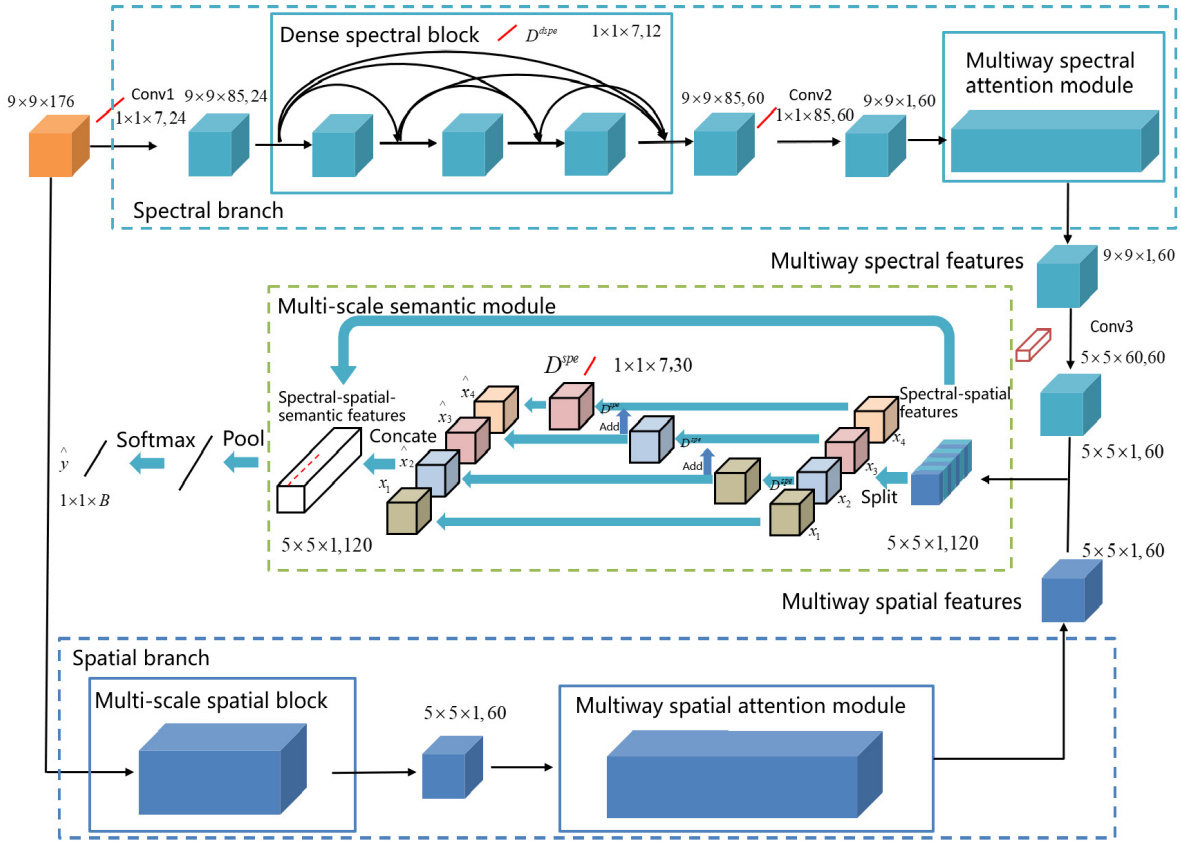


Fig. 3. S^{3}Net with multiway attention mechanism architecture (S^{3}Net). It includes a spectral branch, a spatial branch, and a multiscale semantic module.

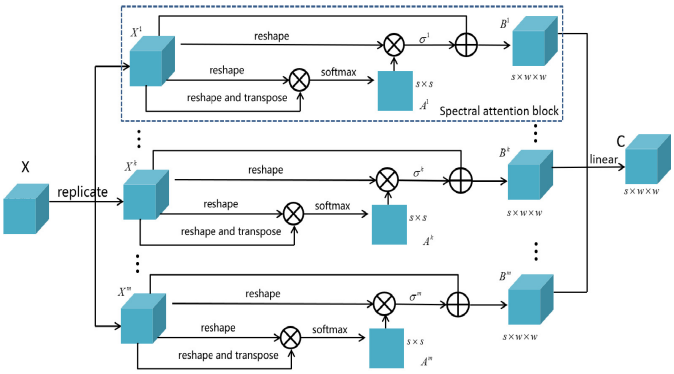


Fig. 4. Proposed multiway spectral attention module contains m way spectral attention blocks.

Finally, these spectral attention maps can generate the multiway spectral feature maps $C \in R^{9\times9\times1,60}$ through linear operation (6). We choose the kth way as an example to explain the spectral attention block, where $k \in [1, m]$.

First, we reshape the 3-D feature cube $X^k \in R^{s\times w\times w}$ into $X^k \in R^{s\times g}$, where $g = w \times w$. Furthermore, the spectral attention map $A^k \in R^{s\times s}$ can be expressed by

$$A^k_{ji} = \frac{\exp\left(X^k_i \times X^k_j\right)}{\sum_{i=1}^{s} \exp\left(X^k_i \times X^k_j\right)} \tag{4}$$

where A^k_{ji} represents the ith channel's influence on the jth channel. X^k_i represents the ith feature map of the X^k. Then, the results of matrix multiplication between A^k and X^k are reshaped into $R^{s\times w\times w}$. Eventually, we use a learnable parameter σ to weight the reshaped results. To preserve the original feature information of X^k, a sum operation between the weighted results and X^k is used to obtain the kth way spectral feature map $B^k \in R^{s\times w\times w}$

$$B^k_j = \sigma \sum_{i=1}^{s} \left(A^k_{ji} X^k_j\right) + X^k_j. \tag{5}$$

We further adopt a linear operation to generate the multiway spectral features C with the size of $s \times w \times w$ from m way spectral attention maps. In this module, $w = 9$, $s = 60$, and

$$C = \sum_{k=1}^{m} \left(F_k B^k + f_k\right) \tag{6}$$

where F_k and f_k represent the weight and bias of the kth spectral feature map B^k. Through the multiway spectral attention module, the key informative features in the spectral features will be highlighted, while the useless spectral features will be suppressed.

B. Spatial Branch Structure With the Multiway Spatial Attention Module

In the S^{3}Net (see Fig. 3), we use the spatial branch to extract the multiway spatial features from the input 3-D cube $9 \times 9 \times 176$. The spatial branch contains a multiscale spatial block (see Fig. 5) and a multiway spatial attention module (see Fig. 6). In the multiscale spatial block, different scale features have strong complementary and related information.

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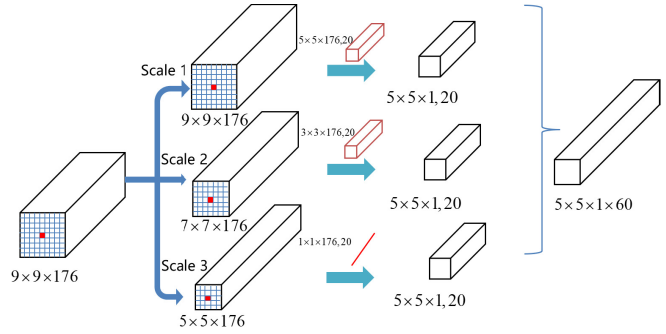


Fig. 5. Multiscale spatial block contained three scales is utilized to learn the multiscale spatial features.

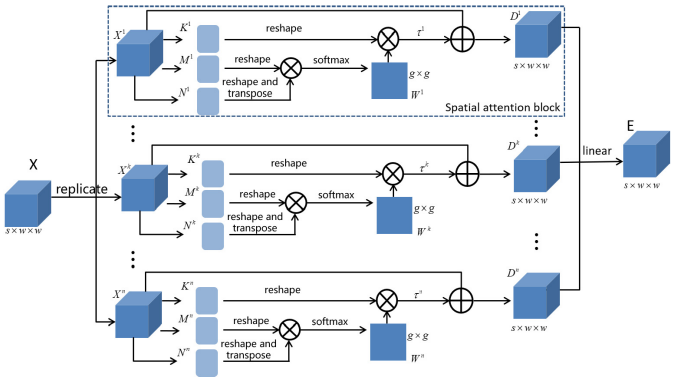


Fig. 6. Proposed multiway spatial attention module contains n way spatial attention blocks.

The multiscale spatial block can extract rich multiscale spatial features.

As shown in Fig. 5, we transform the input 3-D cube $9 \times 9 \times 176$ into three different scales by discarding outermost pixels around its spatial area. The first one is the original 3-D cube. The sizes of the second one and the third one are $7 \times 7 \times 176$ and $5 \times 5 \times 176$ separately. Especially, these three scale 3-D cubes have the same center pixel (the red point in Fig. 5). What is more, we use three different 3-D convolutional layers with the size of $5 \times 5 \times 176, 20$, $3 \times 3 \times 176, 20$, and $1 \times 1 \times 176, 20$ to generate three same size feature maps with the size of $5 \times 5 \times 1, 20$ from the input cube $9 \times 9 \times 176$, $7 \times 7 \times 176$, and $5 \times 5 \times 176$. Finally, the concatenate operation among three feature maps is used to generate the multiscale spatial feature maps with the size of $5 \times 5 \times 1, 60$.

Like the multiway spectral attention module, we also build a novel multiway spatial attention module containing n way spatial attention blocks to refine the multiscale spatial feature maps (see Fig. 6). First, the input 3-D feature cube $X \in R^{5\times5\times1,60}$ is replicated into n copies: $X^1, X^2, \ldots, X^n$. Then, each copy goes through the spatial attention block to generate the spatial feature maps of the same size. Finally, these spatial feature maps can generate the multiway spatial feature maps $D \in R^{5\times5\times1,60}$ through the linear operation (9). We choose the kth way as an example to explain the spatial attention block, where $k \in [1, n]$.

We first use three 3-D convolution layers with the same size of $1 \times 1 \times 60, 60$ to generate new feature maps K^k, M^k, and N^k, respectively. Then, the feature maps K^k, M^k, and N^k are reshaped into the $K^{s\times g}$, $M^{s\times g}$, and $N^{s\times g}$, where $g = w \times w$. Furthermore, a matrix multiplication between K^k and M^k is operated. A softmax layer is attached subsequently to calculate the kth way spatial attention map $W^k \in R^{g\times g}$

$$W_{ji}^k = \frac{\exp\left(K_i^k \times M_j^k\right)}{\sum_{i=1}^{g} \exp\left(K_i^k \times M_j^k\right)} \tag{7}$$

where W_{ji}^k denotes the ith pixel's influence on the jth pixel. K_i^k denotes the ith pixel of the K^k. M_j^k denotes the jth pixel of the M^k. The closer feature representations of the two pixels demonstrate a stronger correlation between them. Then, the results of matrix multiplication between N^k and W^k are reshaped into $R^{s\times w\times w}$. Eventually, we use a learnable parameter τ to weight the reshaped results. A sum operation between the weighted results and N^k is used to obtain the kth way spatial feature maps $G^k \in s \times w \times w$

$$G_j^k = \tau \sum\nolimits_{i=1}^{s} \left(W_{ji}^k N_j^k\right) + N_j^k \tag{8}$$

where N_j^k denotes the jth pixel of N^k.

We further adopt a linear operation to generate the multiway spatial features E with the size of $s \times w \times w$ from n way spatial feature maps. In this module, $w = 5$, $s = 60$, and

$$E = \sum\nolimits_{k=1}^{n} \left(P_k G^k + p_k\right) \tag{9}$$

where P_k and p_k represent the weight and bias of the kth spatial feature map G^k. Through the multiway spatial attention module, the key informative areas in the spatial features will be highlighted, while the useless features will be suppressed.

C. Multiscale Semantic Module

After the spectral branch and the spatial branch, we can obtain the multiway spectral features and the multiway spatial features. To better integrate two types of features rather than simple concatenation operation [33], [34], we first use a 3-D convolutional layer with the size of $5 \times 5 \times 60, 60$ (Conv3) to generate the new multiway spectral features with the shape of $5 \times 5 \times 1, 60$ from the original multiway spectral features with the shape of $9 \times 9 \times 1, 60$ (see Fig. 3). Then, we adopt a cross-fusion operation to obtain spectral–spatial features. The first feature map of the multiway spectral features and the first feature map of the multiway spatial features are concatenated to the first fused feature maps. In the same way, we can obtain the second fused feature maps and the third fused feature maps until the 60th fused feature maps. Finally, these fused feature maps are concatenated to the spectral–spatial feature maps with the shape of $5 \times 5 \times 1, 120$.

To extract more discriminative and representative features for classification (see Fig. 3), we propose a multiscale semantic module. The input spectral–spatial feature maps are first evenly split into four subcubes, denoted by x_1, x_2, x_3, and x_4. x_2 is used to generate the first high-level feature subcube $\hat{x}_2$ through the 3-D convolutional layer D^{spe} that contains 30 convolutional filters of size $1 \times 1 \times 7$. Then, we use the sum

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of the third subcube x_3 and the first high-level feature subcube $\hat{x}_2$ as input to generate the second high-level feature subcube $\hat{x}_3$ via the 3-D convolutional layer D^{spe}. Similarly, we also use the sum of the fourth subcube x_4 and the second high-level feature subcube $\hat{x}_3$ as input to generate the third high-level feature subcube $\hat{x}_4$. To reduce computational burden of the multiscale semantic module, x_1 is kept uncharged. The high-level spectral–spatial feature cube can be obtained through the concatenation operation among $x_1, \hat{x}_2, \hat{x}_3$, and $\hat{x}_4$. Finally, we adopt an identity mapping to directly connect the input spectral–spatial features and the high-level spectral–spatial features and generate more discriminative and representative features for classification. Because these discriminative and representative features have strong semantic information, we name them spectral–spatial–semantic features.

D. S^{3}Net Architecture

In this article, we propose a novel S^{3}Net with the multiway attention mechanism for HSI classification (see Fig. 3), which includes a spectral branch, a spatial branch, and a multiscale semantic module. The spectral branch contains a dense spectral block and a multiway spectral attention module. The spatial branch contains a multiscale spatial block and a multiway spatial attention module. The dense spectral block and the multiscale spatial block can extract the dense spectral features and the multiscale spatial features separately. The multiway spectral attention module and the multiway spatial attention module are adopted to refine and optimize the dense spectral features and multiscale spatial features separately and obtain the multiway spectral features and the multiway spatial features. Then, the cross-fusion operation can generate spectral–spatial features via fusing the two types of features. Finally, the multiscale semantic module extracts the spectral–spatial–semantic features for classification.

Furthermore, we use an average pooling operation (Pool) to extract a $1 \times 1 \times 120$ feature vector from the spectral–spatial–semantic features. Eventually, we adopt a fully connected (FC) layer to generate the predicted value $\hat{y}$ with the size of $1 \times B$, where B represents the number of HSI data set categories. The S^{3}Net prediction process can be expressed as

$$\hat{y} = \text{S}^3\text{Net}\left(x^{r\times r\times L}, \theta\right) \tag{10}$$

where θ represents the parameters of the S^{3}Net. Then, we adopt a softmax regression to obtain the predicted probability, which can be expressed by

$$q_j = \frac{\exp\left(\hat{y}_j\right)}{\sum_{j=1}^{B}\exp\left(\hat{y}_j\right)}. \tag{11}$$

To address the class imbalance problem of the HSI data set, the focal loss [58] is used to calculate the total loss of S^{3}Net

$$L_j^{\text{S}^3\text{Net}} = -y_j\left(1-q_j\right)^r \log\left(q_j\right) \tag{12}$$

where y_j denotes the jth true label belonging to the jth class. The focusing parameter r adjusts the weight balance between well-classified samples (a large number of samples) and hard-classified samples (a small number of samples). Finally, the total loss can be expressed by

$$L^{\text{S}^3\text{Net}} = \frac{1}{M}\sum_{m=1}^{M}\left(\sum_{j=1}^{B} L_j^{\text{S}^3\text{Net}}\right) \tag{13}$$

where M is the size of the minibatch. The RMSprop optimizer [59] is used to minimize the loss $L^{\text{S}^3\text{Net}}$.

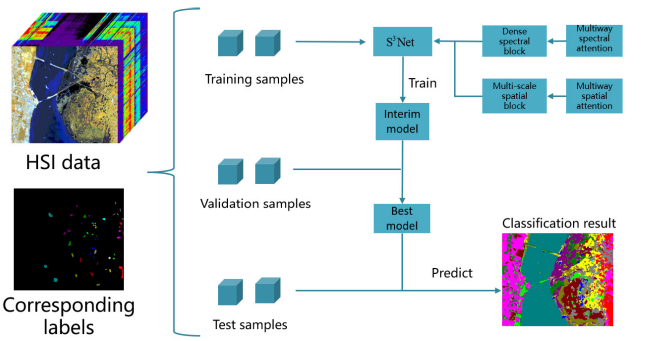


Fig. 7. S^{3}Net framework for HSI classification. Training samples are used to update the S^{3}Net parameters. Validation samples are utilized to monitor the interim model and select the best model. Test samples are employed to evaluate the best model and predict the classification result.

E. Framework of the Proposed Method

Summarizing the above steps, we have proposed a novel framework that can learn the discriminative and representative spectral–spatial–semantic features for HSI classification based on the S^{3}Net. The S^{3}Net framework is shown in Fig. 7.

In the S^{3}Net framework, all available annotated data are separated into three kinds of samples: training, validation, and test samples selected with size $9 \times 9 \times L$. The training and validation samples are utilized to train the S^{3}Net to obtain the best S^{3}Net model. We use the training samples to optimize the parameters of the S^{3}Net model through backpropagating the gradients of the focal loss function (13). The validation samples are used to test the interim model and select the best S^{3}Net model with the highest classification accuracy. Eventually, the best S^{3}Net model is utilized to evaluate the HSI classification performance on the test samples and visualize the classification result (see Fig. 7).

IV. Experiments

In this section, we first introduce four HSI data sets: UP, KSC, PC, and Huoshaoyun data sets. Then, we compare the proposed S^{3}Net with state-of-the-art methods using classification metrics, such as overall accuracy (OA), average accuracy (AA), and kappa coefficient (K). We run experiments ten times with randomly selected training samples based on seeds and report the mean and standard deviation of OA, AA, K, and the classification accuracy of each category.

A. Data Sets' Description

The KSC data contained 13 land-cover types with a wavelength scope of 400–2500 nm. It had 512×614 pixels

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TABLE I
CATEGORY INFORMATION FOR THE UP DATA SET

Order Number	Classification	Number of Samples
1	Asphalt	6631
2	Meadows	18649
3	Gravel	2099
4	Trees	3064
5	Painted metal sheets	1345
6	Bare soil	5029
7	Bitumen	1330
8	Self-blocking bricks	3692
9	Shadows	947
	Total	42776

TABLE II
CATEGORY INFORMATION FOR THE KSC DATA SET

Order Number	Classification	Number of Samples
1	Scrub	347
2	Willow swamp	243
3	CP hammock	256
4	Slash pine	252
5	Oak/broadleaf	161
6	Hardwood	229
7	Swamp	105
8	Graminoid marsh	390
9	Spartina marsh	520
10	Cattail marsh	404
11	Salt marsh	419
12	Mud flats	503
13	Water	927
	Total	5211

TABLE III
CATEGORY INFORMATION FOR THE PC DATA SET

Order Number	Classification	Number of Samples
1	Waters	65971
2	Trees	7598
3	Asphalt	3090
4	Self-Blocking Bricks	2685
5	Bitumen	6584
6	Tiles	9248
7	Shadows	7287
8	Meadows	42826
9	Bare soil	2863
	Total	148152

with 176 bands. The UP data contained nine urban land-cover classes with a wavelength scope of 430–860 nm. It had 610 × 340 pixels with 103 bands. The PC data had 1096 × 715 samples with 102 bands, which contained nine urban land-cover classes.

The Huoshaoyun data were collected by GF-5 satellite and had 2057 × 1975 pixels, which contained 11 lithology classes. We used 164 bands with a wavelength scope of 390–2513 nm to collect the HSI from the Huoshaoyun area in Xinjiang province, China. Huoshaoyun data are manually annotated, and the experts have explored the lithology classes that are helpful for mineral exploration at the scene. Tables I, II, III, and IV summarize the categories and numbers of samples for UP, KSC, PC, and Huoshaoyun data sets separately. It can be found that all categories in the four data sets have class imbalance problems. Take the UP data set as an example, the Meadows class has 18 649 samples, whereas the Shadows has only 947 samples.

TABLE IV
CATEGORY INFORMATION FOR THE HUOSHAOYUN DATA SET

Order Number	Classification	Number of Samples
1	Sandstone-1	7785
2	Sandstone-2	6137
3	Carbonation-1	8561
4	Carbonation-2	10569
5	Snowy water-1	19984
6	Sandstone-3	10267
7	Sandstone-4	6916
8	Snowy water-2	1264
9	Snowy water-3	4819
10	Chloritization-1	2239
11	Chloritization-2	1394
	Total	79935

B. Framework Setting

1) Basic Settings: In this article, the proposed S^{3}Net framework is implemented with Torch. After designing the S^{3}Net framework, we configure some important parameters. All data sets are randomly divided into the training, validation, and test samples from ten random seeds. Notably, the number of validation samples is the same as that of training samples. The batch size is set to 16, and the RMSprop optimizer [59] is used for stochastic optimization. The number of training epochs is set to 200, and the learning rate is set to 0.0005.

2) Number Setting of the Spectral Attention Blocks and the Spatial Attention Blocks: The number of spectral attention blocks determines how many subspaces the multiway spectral attention module needs. Similarly, the number of spatial attention blocks determines how many subspaces the multiway spatial attention module needs. It is very necessary to adopt the appropriate numbers of the spectral attention blocks and the spatial attention blocks to refine the extracted spectral features and the multiscale spatial features separately. As shown in Fig. 8, we use the heat map to display the OA of the S^{3}Net on the different numbers of spectral attention blocks and spatial attention blocks on four HSI data sets. On the UP data set, the S^{3}Net achieves the highest OA on two spatial attention blocks and five spectral attention blocks. On the KSC data set, the S^{3}Net achieves the highest OA on three spatial attention blocks and seven spectral attention blocks. On the PC data set, the S^{3}Net achieves the highest OA on two spatial attention blocks and three spectral attention blocks. On the Huoshaoyun data set, the S^{3}Net achieves the highest OA on three spatial attention blocks and three spectral attention blocks. These experimental results demonstrate that the multiple spectral attention blocks and the multiple spatial attention blocks in the S^{3}Net are extremely effective for HSI classification.

3) Focusing Parameter Setting of the Focal Loss Function: The focal loss function with a focusing parameter r is used to train the S^{3}Net framework. Particularly, the focusing parameters $r = 1, 1.5, 2, 2.5, 3$, and 3.5 are used in the UP, KSC, PC, and Huoshaoyun data sets. As shown in Fig. 9, on the UP data set [see Fig. 9(a)], the S^{3}Net has the highest OA, AA, and K at $r = 3$. On the KSC data set [see Fig. 9(b)], the S^{3}Net has the highest OA, AA, and K at $r = 2.5$. On the PC data set [see Fig. 9(c)], the S^{3}Net has the highest OA, AA, and K at $r = 2$. On the Huoshaoyun data set [see Fig. 9(d)], and the S^{3}Net has

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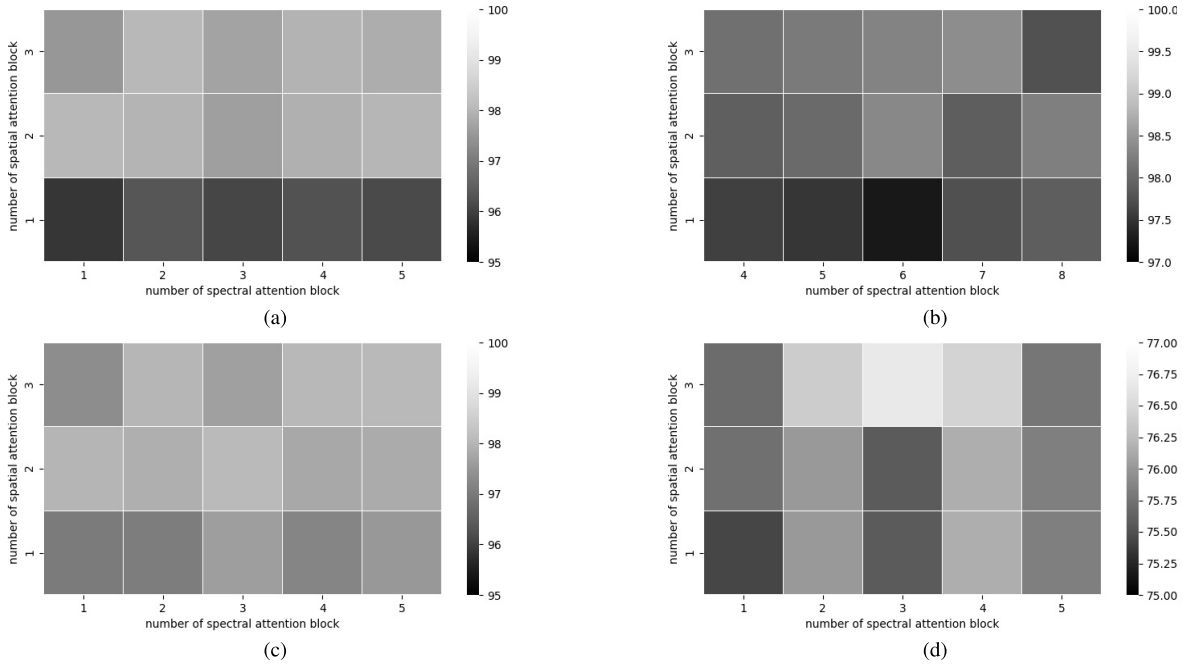


Fig. 8. OA for the S^{3}Net with different numbers of the spectral attention blocks and the spatial attention blocks on four HSI data sets: (a) UP data set; (b) KSC data set; (c) PC data set; and (d) Huoshaoyun data set.

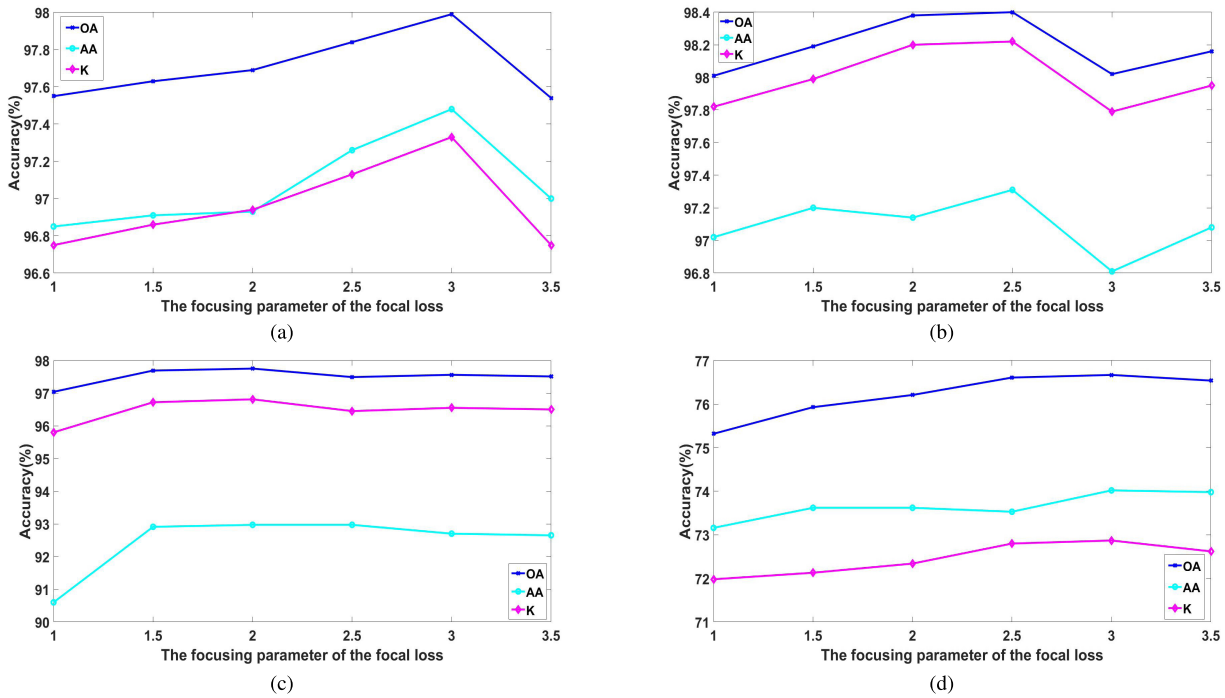


Fig. 9. OA, AA, and K for different focusing parameters of the focal loss in four HSI data sets: (a) UP data set; (b) KSC data set; (c) PC data set; and (d) Huoshaoyun data set.

the highest OA, AA, and K at $r = 3$. In general, the S^{3}Net has higher OA, AA, and K on the focal loss function with the appropriate focusing parameters ($r = 3, r = 2.5, r = 2$, and $r = 3$) than the standard cross entropy loss ($r = 1$) on the UP, KSC, PC, and Huoshaoyun data sets. These experimental results demonstrate that the focal loss function is very useful for the class imbalance problem.

C. Classification Maps and Categorized Results

We compare the proposed S^{3}Net framework with other deep learning-based methods, such as a contextual deep CNN (CDCNN) [10], SSRN [15], FDSSC [41], DBMA [33], and DBDA [34]. Since these methods and the proposed S^{3}Net are all 3-D CNN-based methods, their input spatial sizes are fixed at 9×9 for a fair comparison. Tables V, VI, VII, and VIII

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TABLE V
OA, AA, AND K FOR EACH HSI CATEGORY IN THE UP DATA SET. DATA ARE GIVEN AS MEAN ± STANDARD DEVIATION

Methods	CDCNN	SSRN	FDSSC	DBMA	DBDA	S^3Net
OA (%)	87.83 ± 1.26	95.93 ± 0.87	94.93 ± 0.95	94.72 ± 1.74	96.30 ± 0.95	**98.03 ± 0.56**
AA (%)	86.36 ± 1.54	94.85 ± 1.27	94.68 ± 1.15	95.49 ± 2.01	95.06 ± 1.21	**97.51 ± 0.64**
K (%)	83.81 ± 1.61	94.59 ± 1.16	92.57 ± 1.25	92.95 ± 2.01	95.10 ± 1.28	**97.39 ± 0.75**
1	86.30 ± 3.14	96.97 ± 2.02	96.88 ± 1.98	94.86 ± 1.68	94.64 ± 3.45	**98.36 ± 1.08**
2	93.51 ± 1.70	98.27 ± 0.82	97.57 ± 1.11	96.57 ± 1.01	99.24 ± 0.47	**99.42 ± 0.30**
3	61.26 ± 8.29	82.10 ± 9.95	89.97 ± 8.62	**100 ± 0.00**	90.75 ± 6.85	97.54 ± 3.24
4	96.16 ± 2.73	98.81 ± 1.34	**99.22 ± 1.21**	97.44 ± 1.38	96.58 ± 2.41	98.60 ± 0.82
5	98.29 ± 2.00	99.46 ± 1.09	**99.55 ± 1.01**	95.69 ± 0.95	98.92 ± 1.92	98.96 ± 1.32
6	79.75 ± 6.98	97.76 ± 1.71	97.97 ± 0.89	96.78 ± 0.82	97.80 ± 1.75	**98.11 ± 1.46**
7	83.92 ± 9.95	96.60 ± 4.69	**100 ± 0.00**	95.69 ± 3.21	94.35 ± 4.59	97.74 ± 5.61
8	81.97 ± 4.21	84.94 ± 6.17	70.97 ± 4.26	78.93 ± 3.62	86.73 ± 5.58	**90.59 ± 5.73**
9	96.08 ± 2.54	98.74 ± 1.68	**100 ± 0.00**	99.55 ± 0.73	96.51 ± 3.08	97.67 ± 1.91

TABLE VI
OA, AA, AND K FOR EACH HSI CATEGORY IN THE KSC DATA SET. DATA ARE GIVEN AS MEAN ± STANDARD DEVIATION

Methods	CDCNN	SSRN	FDSSC	DBMA	DBDA	S^3Net
OA (%)	81.77 ± 4.89	93.67 ± 1.63	93.97 ± 2.19	93.36 ± 1.91	96.87 ± 1.43	**98.40 ± 0.59**
AA (%)	71.03 ± 8.95	90.45 ± 3.24	91.05 ± 4.32	89.72 ± 2.79	94.88 ± 1.93	**97.22 ± 1.09**
K (%)	79.69 ± 5.46	92.96 ± 1.81	93.28 ± 2.45	92.61 ± 2.13	96.51 ± 1.59	**98.21 ± 0.66**
1	96.79 ± 2.54	99.06 ± 1.19	97.89 ± 4.17	99.74 ± 0.55	99.88 ± 0.26	**100 ± 0.00**
2	64.28 ± 21.50	93.08 ± 7.35	91.94 ± 7.74	95.16 ± 5.40	96.80 ± 4.26	**98.61 ± 2.57**
3	50.74 ± 24.73	70.42 ± 28.35	70.71 ± 29.94	73.23 ± 12.67	86.49 ± 7.50	**95.70 ± 3.93**
4	36.67 ± 14.30	62.70 ± 12.38	69.35 ± 19.12	70.79 ± 8.30	78.23 ± 14.04	**87.55 ± 5.11**
5	25.67 ± 18.06	71.94 ± 21.73	67.86 ± 28.38	69.91 ± 17.39	83.14 ± 11.72	**91.29 ± 7.58**
6	68.89 ± 11.80	96.80 ± 4.73	94.39 ± 10.54	92.10 ± 8.98	98.19 ± 2.20	**98.46 ± 3.85**
7	49.69 ± 40.69	89.87 ± 8.62	**95.49 ± 6.80**	79.20 ± 13.57	93.27 ± 7.34	94.74 ± 8.17
8	69.70 ± 12.11	96.22 ± 3.64	98.50 ± 1.44	93.86 ± 4.09	**99.25 ± 1.15**	98.98 ± 1.46
9	84.40 ± 6.50	98.95 ± 1.73	99.11 ± 2.25	96.75 ± 4.43	**100 ± 0.00**	99.87 ± 0.31
10	86.54 ± 11.72	99.28 ± 2.15	99.87 ± 0.39	99.42 ± 0.93	**100 ± 0.00**	**100 ± 0.00**
11	98.43 ± 1.59	98.71 ± 1.35	**99.45 ± 0.93**	99.13 ± 1.73	98.88 ± 1.73	99.41 ± 0.95
12	92.29 ± 6.37	99.14 ± 1.52	99.27 ± 1.07	97.09 ± 3.11	**99.32 ± 0.69**	99.29 ± 0.99
13	99.30 ± 0.81	99.72 ± 0.84	99.83 ± 0.50	**100 ± 0.00**	**100 ± 0.00**	**100 ± 0.00**

TABLE VII
OA, AA, AND K FOR EACH HSI CATEGORY IN THE PC DATA SET. DATA ARE GIVEN AS MEAN ± STANDARD DEVIATION

Methods	CDCNN	SSRN	FDSSC	DBMA	DBDA	S^3Net
OA (%)	95.36 ± 1.56	97.54 ± 1.47	97.53 ± 1.34	96.96 ± 1.07	97.33 ± 0.92	**98.08 ± 0.51**
AA (%)	85.70 ± 5.17	92.39 ± 3.71	92.35 ± 3.24	90.82 ± 4.11	91.87 ± 3.11	**93.93 ± 1.75**
K (%)	93.40 ± 2.28	96.53 ± 2.08	96.51 ± 1.89	95.70 ± 1.53	96.22 ± 0.91	**97.29 ± 0.73**
1	99.07 ± 1.50	**99.99 ± 0.03**	99.97 ± 0.06	99.75 ± 0.37	99.83 ± 0.27	99.93 ± 0.09
2	89.53 ± 5.48	96.78 ± 3.05	96.84 ± 2.38	94.74 ± 3.85	94.56 ± 4.88	**97.61 ± 1.98**
3	**81.68 ± 8.22**	77.49 ± 12.95	76.13 ± 12.62	81.21 ± 9.12	77.88 ± 11.69	78.20 ± 8.68
4	58.27 ± 12.34	73.91 ± 21.25	72.27 ± 16.06	66.39 ± 23.15	74.78 ± 13.13	**83.19 ± 13.01**
5	83.46 ± 10.42	93.56 ± 4.05	**96.35 ± 4.57**	90.12 ± 10.57	92.59 ± 5.77	95.27 ± 3.19
6	89.99 ± 4.09	93.55 ± 6.79	**95.37 ± 3.33**	90.98 ± 4.24	93.71 ± 2.41	94.35 ± 3.26
7	91.95 ± 6.98	97.57 ± 2.99	95.66 ± 6.47	97.93 ± 3.27	97.59 ± 1.59	**98.09 ± 2.73**
8	98.20 ± 2.29	**99.90 ± 0.08**	99.89 ± 0.11	99.03 ± 0.79	99.22 ± 0.65	99.68 ± 0.19
9	79.14 ± 27.51	98.71 ± 1.78	98.63 ± 2.78	97.23 ± 3.39	96.71 ± 3.92	**99.06 ± 0.98**

show the results of the OA, AA, K, and accuracy of each class on UP, KSC, PC, and Huoshaoyun data sets separately, and the best accuracies are shown in bold.

1) Classification Maps and Categorized Results for Three Widely Used UP, KSC, and PC Data Sets: On the UP data set, since it has sufficient samples, there are enough samples for each class even if we just choose 0.5% training samples. As shown in Table V, the proposed S^3Net obtains the higher values of OA, AA, and K than other methods. Our model achieves the best accuracies in class 1, 2, 6, and 8 of all categories. Especially, the S^3Net achieves the 3.72% (class 1) and 3.86% (class 8) improvements than the second highest value. Meanwhile, the S^3Net achieves the comparable accuracies for FDSSC in classes 4 and 5. Fig. 10(a)–(g) shows the classification maps of the ground truth, CDCNN, SSRN, FDSSC, DBMA, DBDA, and ours, respectively. In general, the classification map of the S^3Net yields a smoother visual effect and shows fewer misclassifications than other compared methods. To better compare the classification results, we display one zoom region in the middle of these classification maps with the special classes 1, 2, 6, and 8 [see Fig. 10(h)–(n)]. It is obvious that the result [see Fig. 10(n)] of our method is cleaner and has fewer misclassifications than other methods [see Fig. 10(i)–(m)].

On the KSC data set, as shown in Table VI, the proposed S^3Net obtains the higher values of OA, AA, and K than other methods on the 5% training samples. Our model achieves the best accuracies in classes 1, 2, 3, 4, 5, 6, 10, and 13 of

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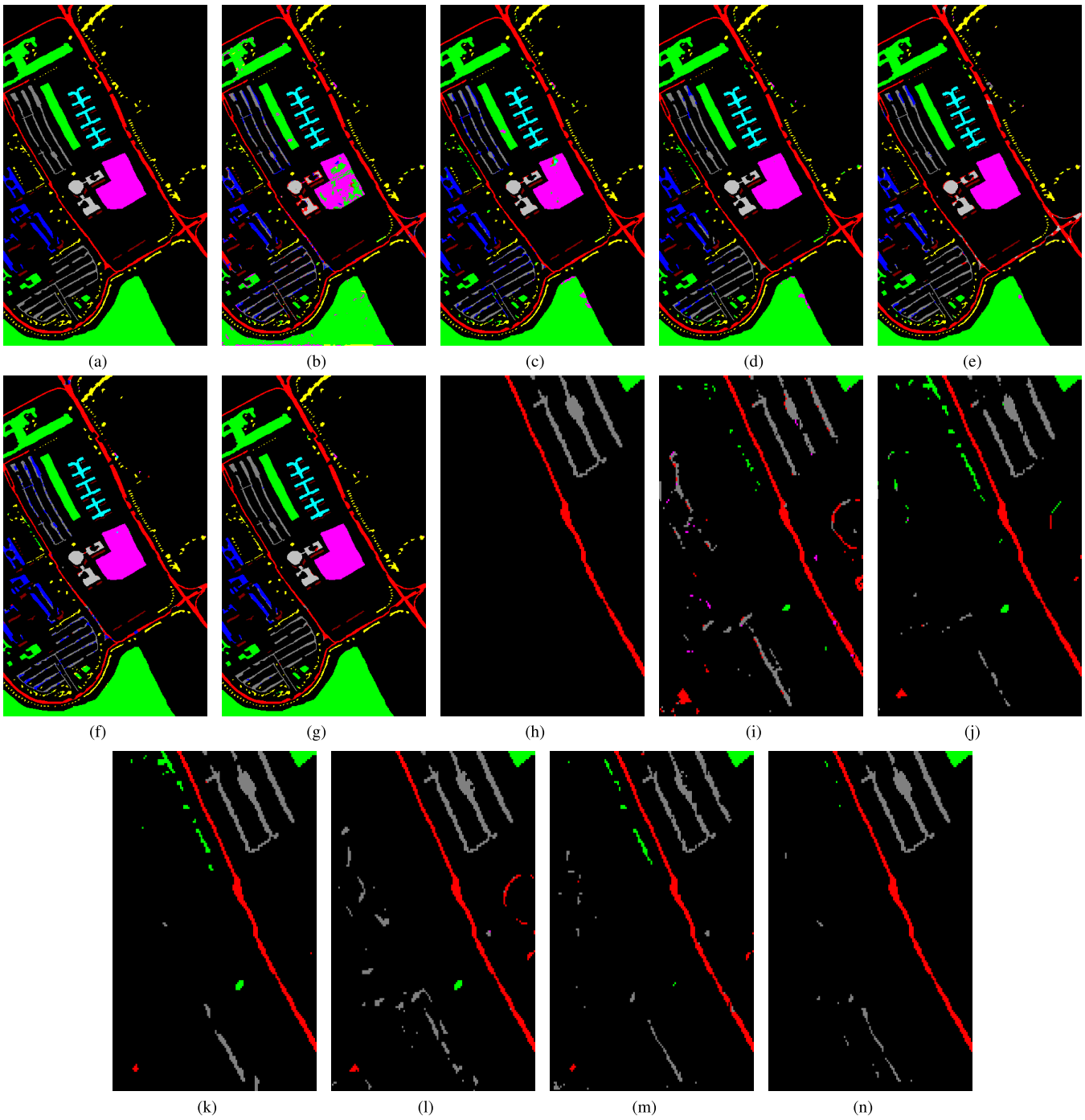


Fig. 10. Classification maps for the UP data set obtained by different methods. (a) Ground-truth map. (b) CDCNN. (c) SSRN. (d) FDSSC. (e) DBMA. (f) DBDA. (g) Our model. (h) Zoomed-in view special classes of ground-truth map. (i) Zoomed-in view special classes of CDCNN. (j) Zoomed-in view special classes of SSRN. (k) Zoomed-in view special classes of FDSSC. (l) Zoomed-in view special classes of DBMA. (m) Zoomed-in view special classes of DBDA. (n) Zoomed-in view special classes of our model.

all categories. Especially, the S^{3}Net achieves 1.81% (class 2), 9.21% (class 3), 9.32% (class 4), and 8.15% (class 5) improvement than the second highest value. Meanwhile, the S^{3}Net achieves comparable accuracies for DBDA in classes 9 and 12. It means that our model can capture more distinctive features between different classes than other methods. Fig. 11(a)–(g) shows the classification maps of the ground truth, CDCNN, SSRN, FDSSC, DBMA, DBDA, and ours, respectively. To better compare the classification results, we display one zoom region in the upper right corner of these classification maps with the special classes 1, 2, 3, 4, 5, and 6 [see Fig.11(h)–(n)]. It is obvious that our method [see Fig. 11(n)] has fewer misclassifications than other methods [see Fig. 11(i)–(m)].

On the PC data set, as shown in Table VII, the proposed S^{3}Net obtains the higher values of OA, AA, and K than other

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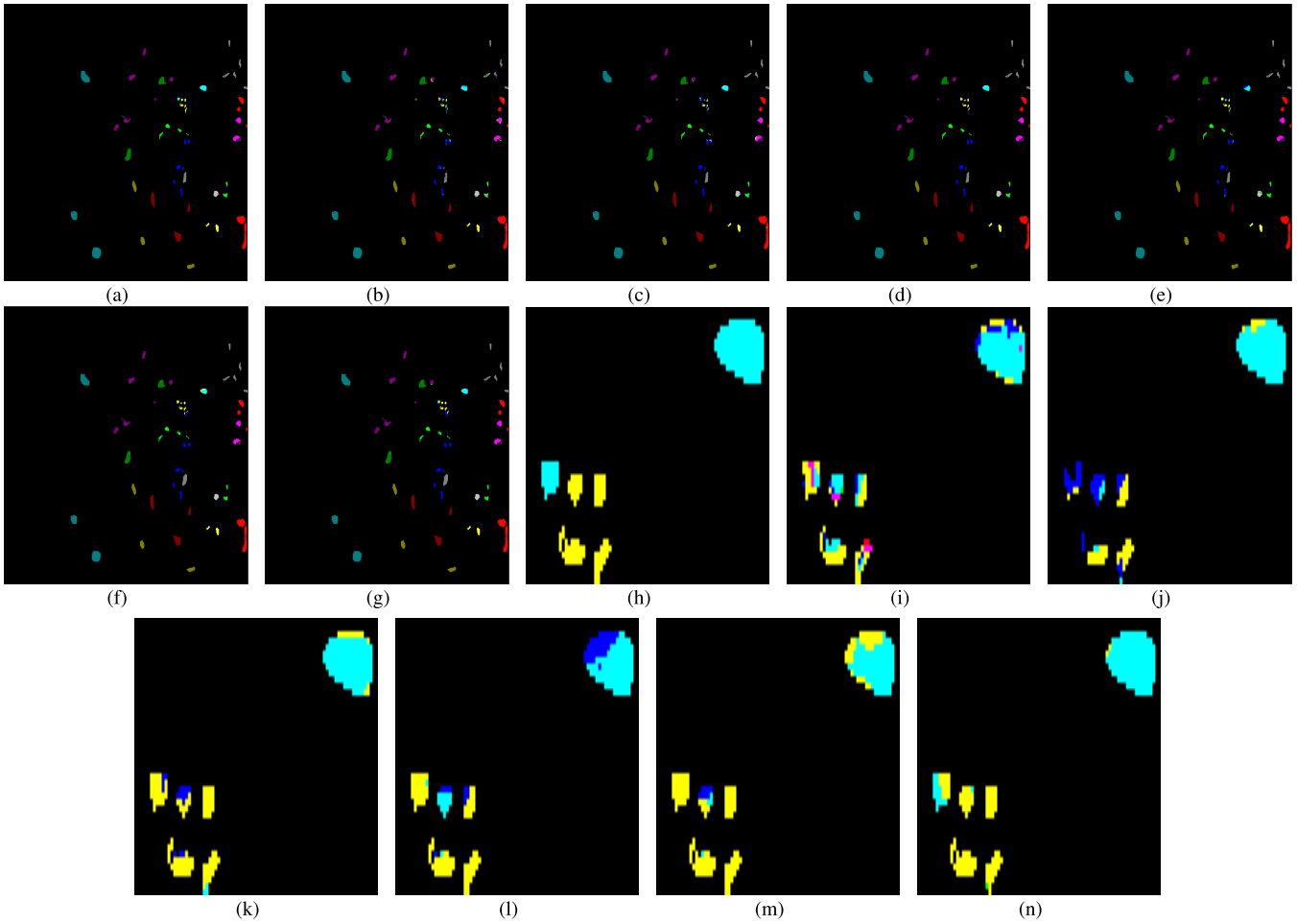


Fig. 11. Classification maps for the KSC data set obtained by different methods. (a) Ground-truth map. (b) CDCNN. (c) SSRN. (d) FDSSC. (e) DBMA. (f) DBDA. (g) our model. (h) Zoomed-in view special classes of ground-truth map. (i) Zoomed-in view special classes of CDCNN. (j) Zoomed-in view special classes of SSRN. (k) Zoomed-in view special classes of FDSSC. (l) Zoomed-in view special classes of DBMA. (m) Zoomed-in view special classes of DBDA. (n) Zoomed-in view special classes of our model.

TABLE VIII

OA, AA, AND K FOR EACH HSI CATEGORY IN THE HUOSHAOYUN DATA SET. DATA ARE GIVEN AS MEAN ± STANDARD DEVIATION

Methods	CDCNN	SSRN	FDSSC	DBMA	DBDA	S^3Net
OA (%)	67.42 ± 2.97	74.40 ± 3.13	72.63 ± 2.11	71.17 ± 2.01	72.25 ± 2.43	**76.67 ± 1.84**
AA (%)	63.83 ± 2.85	71.82 ± 2.56	70.65 ± 2.55	68.48 ± 2.14	69.49 ± 2.33	**74.02 ±1.78**
K (%)	62.13 ± 3.50	70.37 ± 3.56	68.09 ± 2.43	66.63 ± 2.39	67.87 ± 2.78	**72.87 ± 2.16**
1	60.90 ± 12.28	77.15 ± 8.40	73.98 ± 11.21	75.20 ± 8.10	78.02 ± 7.46	**78.64 ± 10.23**
2	56.83 ± 10.29	82.84 ± 8.83	84.22 ± 9.61	**85.15 ± 6.31**	77.61 ± 7.99	80.17 ± 6.85
3	66.94 ± 5.01	72.03 ± 8.61	76.83 ± 8.99	72.70 ± 5.98	75.94 ± 5.88	**78.60 ± 6.48**
4	67.97 ± 10.17	76.70 ± 4.26	77.53 ± 6.51	72.83 ± 9.09	75.97 ± 5.58	**79.62 ± 4.38**
5	79.76 ± 3.37	**82.40 ± 4.26**	75.56 ± 3.02	80.55 ± 3.95	79.34 ± 2.19	81.12 ± 2.87
6	74.66 ± 7.42	76.71 ± 10.76	73.72 ± 7.11	69.73 ± 7.49	71.92 ± 8.46	**78.66 ± 5.71**
7	64.85 ± 9.11	66.02 ± 8.33	60.07 ± 11.89	56.31 ± 9.20	55.40 ± 8.70	**66.99 ± 6.12**
8	54.77 ± 15.25	79.12 ± 11.15	**81.13 ± 8.75**	76.62 ± 13.19	80.40 ± 8.73	79.79 ± 8.61
9	51.81 ± 11.43	55.33 ± 10.60	56.41 ± 11.39	51.76 ± 15.26	54.12 ± 12.74	**60.82 ± 9.84**
10	64.81 ± 8.55	62.35 ± 9.19	60.61 ± 6.53	63.33 ± 8.79	63.48 ± 10.02	**68.74 ± 9.22**
11	58.86 ± 12.42	59.40 ± 11.21	57.14 ± 12.39	49.09 ± 5.45	52.23 ± 6.94	**61.07 ± 14.37**

methods on the 0.1% training samples. Our model achieves the best accuracies in classes 2, 4, 7, and 9 of all categories. Especially, the S^3Net achieves 0.77% (class 2), 8.41% (class 4), and 0.35% (class 9) improvements than the second highest value. Meanwhile, the S^3Net achieves the comparable accuracies for SSRN in classes 1 and 8. Fig. 12(a)–(g) shows the classification maps of the ground truth, CDCNN, SSRN, FDSSC, DBMA, DBDA, and ours, respectively. To better compare the classification results, we display one zoom region in the middle of these classification maps with the classes 2, 4, 7, and 9 [see Fig. 12(h)–(n)]. It can be found that our method [see Fig. 12(n)] has fewer misclassifications than other

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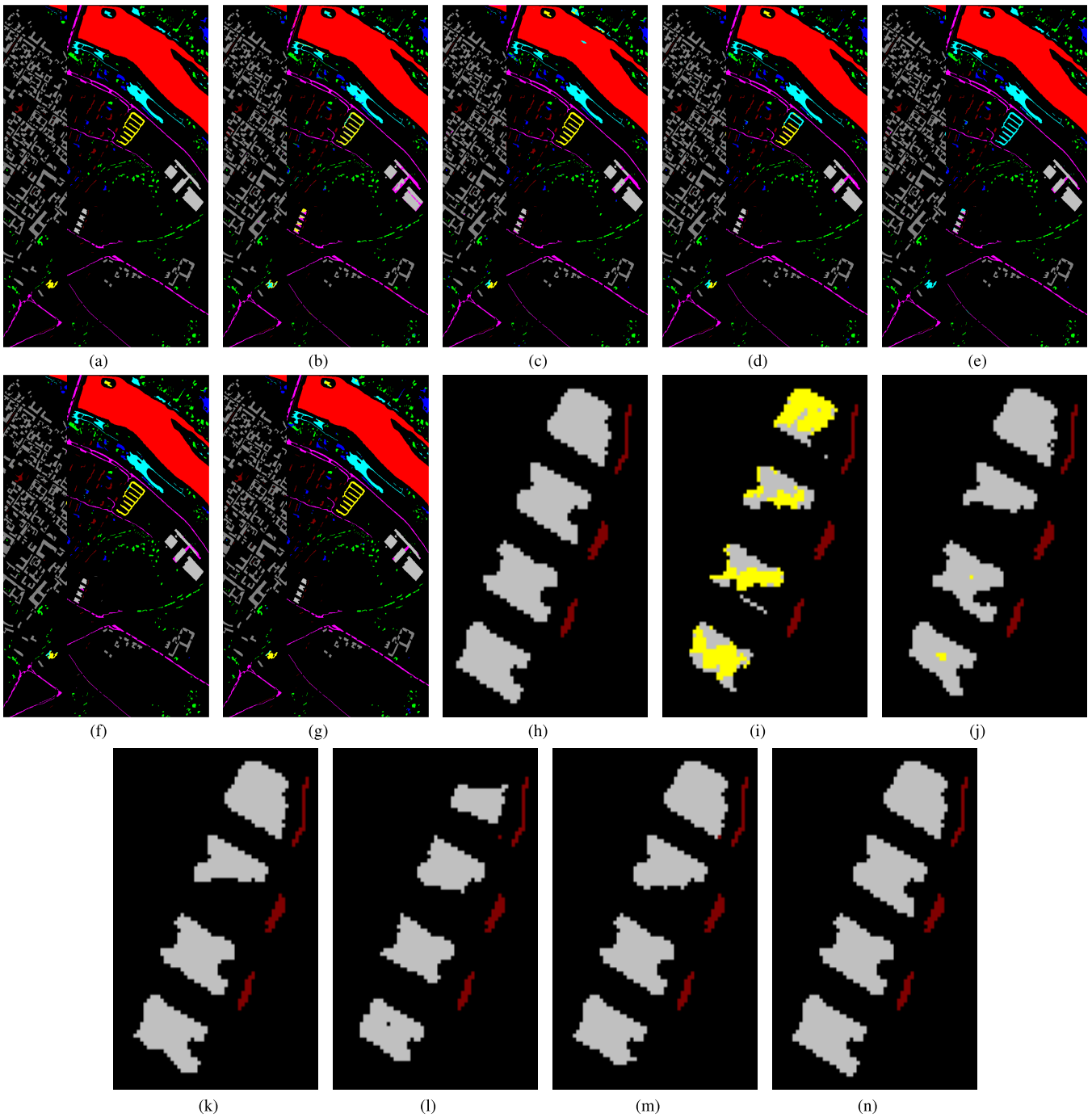


Fig. 12. Classification maps for the PC data set obtained by different methods. (a) Ground-truth map. (b) CDCNN. (c) SSRN. (d) FDSSC. (e) DBMA. (f) DBDA. (g) Our model. (h) Zoomed-in view special classes of ground-truth map. (i) Zoomed-in view special classes of CDCNN. (j) Zoomed-in view special classes of SSRN. (k) Zoomed-in view special classes of FDSSC. (l) Zoomed-in view special classes of DBMA. (m) Zoomed-in view special classes of DBDA. (n) Zoomed-in view special classes of our model.

methods [see Fig. 12(i)–(m)]. Remarkably, the shape of our zoom region classification map of special class [see Fig. 12(n)] is the closest to the ground truth [see Fig. 12(h)].

2) Classification Maps and Categorized Results for the Huoshaoyun Data Set: Since the Huoshaoyun data set has sufficient samples, we adopt the limited training samples (0.1%) for each class to train the S^{3}Net. As shown in Table VIII, the S^{3}Net obtains the higher values of OA, AA, and K than other methods. Our model achieves the best accuracies in classes 1, 3, 4, 5, 6, 9, 10, and 11 of all categories. Especially, the S^{3}Net achieves 1.97% (class 3), 2.09% (class 4), 1.95% (class 6), 4.41% (class 9), 3.93% (class 10), and 1.67% (class 11) improvements than the second highest value. Fig. 13(a)–(g) shows the classification maps of the ground truth, CDCNN, SSRN, FDSSC, DBMA, DBDA, and ours, respectively. To better compare the classification

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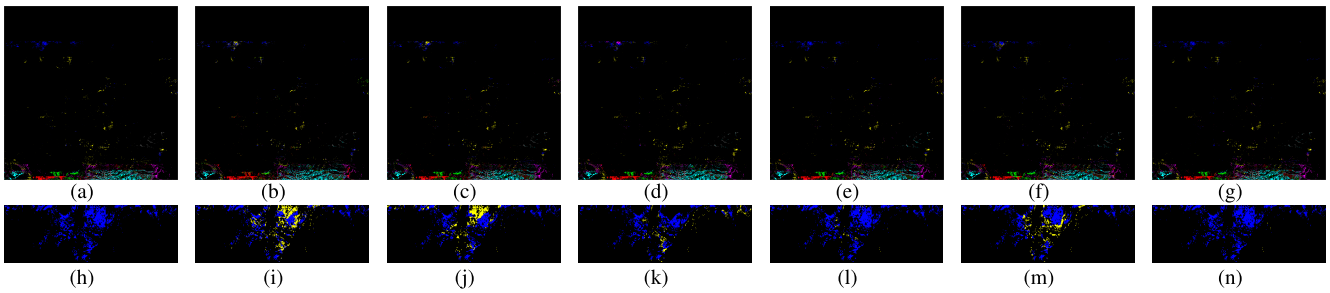


Fig. 13. Classification maps for the Huoshaoyun data set obtained by different methods. (a) Ground-truth map. (b) CDCNN. (c) SSRN. (d) FDSSC. (e) DBMA. (f) DBDA. (g) Our model. (h) Zoomed-in view special classes of ground-truth map. (i) Zoomed-in view special classes of CDCNN. (j) Zoomed-in view special classes of SSRN. (k) Zoomed-in view special classes of FDSSC. (l) Zoomed-in view special classes of DBMA. (m) Zoomed-in view special classes of DBDA. (n) Zoomed-in view special classes of our model.

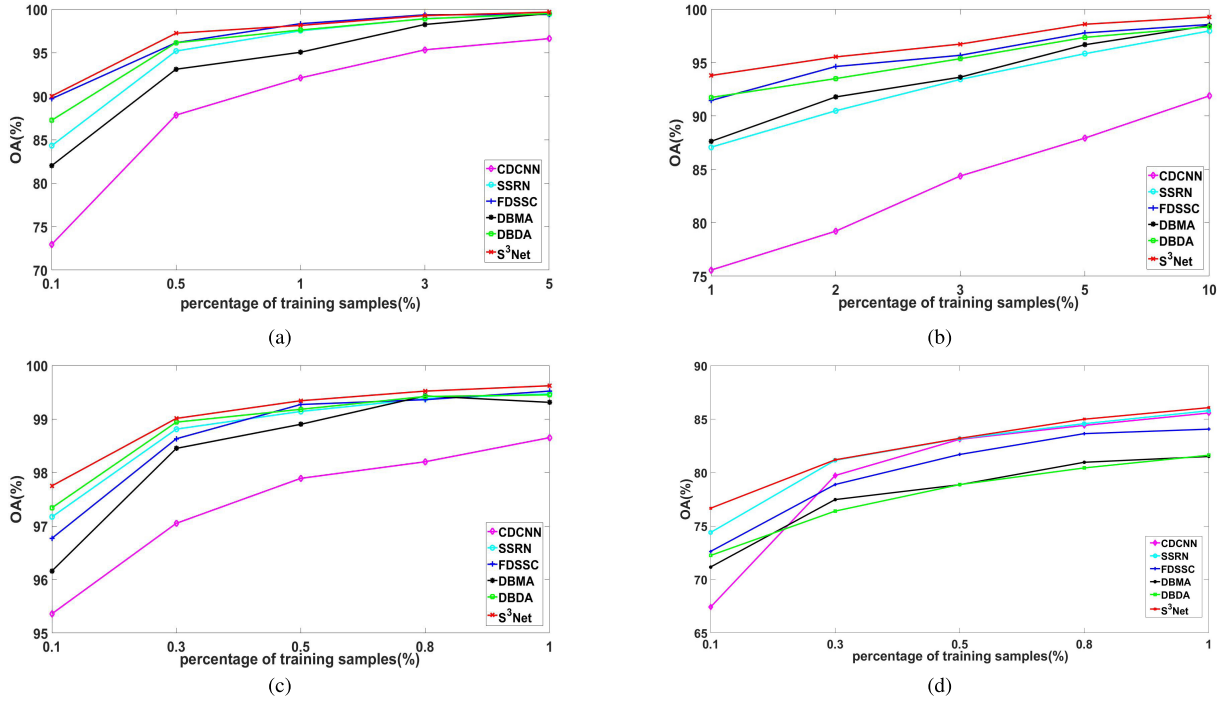


Fig. 14. OA of different methods for different percentage of training samples on four HSI data sets: (a) UP data set; (b) KSC data set; (c) PC data set; and (d) Huoshaoyun data set.

results, we display one zoom region in the top left corner of these classification maps with the classes 3, 4, 9, and 10 [see Fig. 13(h)–(n)]. It can be found that our method [see Fig. 13(n)] is cleaner and has fewer misclassifications than other methods [see Fig. 13(i)–(m)].

The above experimental results demonstrate that our proposed S^{3}Net obtains the best performance in terms of OA, AA, and K on three widely used data sets and the new Huoshaoyun data set.

To further demonstrate the robustness and generalizability of the proposed S^{3}Net toward different numbers of training samples, we conduct lots of experiments among CDCNN [10], SSRN [15], FDSSC [41], DBMA [42], DBDA [34], and our S^{3}Net on different numbers of training samples. Fig. 14 displays the OAs of different methods with different numbers of training samples on the UP, KSC, PC, and Huoshaoyun data sets. In detail, on the UP data set [see Fig. 14(a)], we use 0.1%, 0.5%, 1%, 3%, and 5% of the samples as training sets, respectively. Our proposed S^{3}Net achieves higher OA than other methods on the very limited 0.1%, 0.5% UP training samples. When the percentage of training samples is higher than or equal to 1%, the OA of S^{3}Net is similar to that of DBDA and FDSSC. This is simply because the classification accuracy is very high (higher than 98% OA). On the KSC data set [see Fig. 14(b)], we use 1%, 2%, 3%, 5%, and 10% of the samples as training sets, respectively. Our proposed S^{3}Net achieves higher OA than other methods on all different percentage of training samples. On the PC [see Fig. 14(c)] and Huoshaoyun [see Fig. 14(d)] data sets, we use 0.1%, 0.3%, 0.5%, 0.8%, and 1% of the samples as training sets, respectively. Our proposed S^{3}Net also achieves higher OA than other methods on all different percentage of training samples.

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TABLE IX

OA, AA, AND K FOR DIFFERENT MODULES IN THE S^{3}NET ON THE KSC DATA SET. DATA ARE GIVEN AS MEAN ± STANDARD DEVIATION

Models	OA (%)	AA (%)	K (%)
S^{3}Net without multi-scale semantic module	97.34 ± 0.71	95.30 ± 1.12	97.04 ± 0.78
S^{3}Net without spatial branch	96.86 ± 3.53	96.04 ± 2.28	96.51 ± 3.92
S^{3}Net without spectral branch	97.26 ± 1.01	94.97 ± 1.78	96.95 ± 1.12
S^{3}Net	**98.40 ± 0.59**	**97.22 ± 1.09**	**98.21 ± 0.66**

These experimental results demonstrate that our proposed S^{3}Net has stronger robustness and generalizability than other methods.

D. Ablation Experiments About Each Module in the S^{3}Net

To further evaluate the effectiveness of each module in the S^{3}Net, we perform an ablation experiment to show the OA, AA, and K for different modules on the KSC data set.

As shown in Table IX, the experimental results demonstrate that our proposed S^{3}Net obtains the best performance in terms of OA, AA, and K than other methods. In particular, the OA, AA, and K of the S^{3}Net without multiscale semantic module, S^{3}Net without spatial branch, and S^{3}Net without spectral branch are 1%–2.3% lower than those of the S^{3}Net model. This illustrates that each module in the S^{3}Net works together to achieve better performance.

V. CONCLUSION

In this article, we propose a novel S^{3}Net with the multiway attention mechanism for HSI classification. The S^{3}Net consists of a spectral branch, a spatial branch, and a multiscale semantic module. The spectral branch can learn the multiway spectral features with a dense spectral block and a multiway spectral attention module. The spatial branch can learn the multiway spatial features with a multiscale spatial block and a multiway spatial attention module. Furthermore, the multiway spectral features and the multiway spatial features are fused to generate the spectral–spatial features via the cross-fusion operation. Finally, the multiscale semantic module is proposed to extract spectral–spatial–semantic features that are used for classification. A large number of experimental results show that, on the UP, KSC, and PC, the proposed S^{3}Net achieves higher accuracy than state-of-the-art methods on the limited training samples. Remarkably, the proposed method is also used on the mineral exploration Huoshaoyun data set and achieves the best performance.

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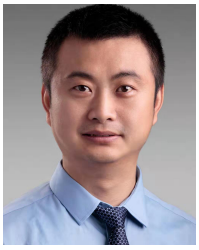
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